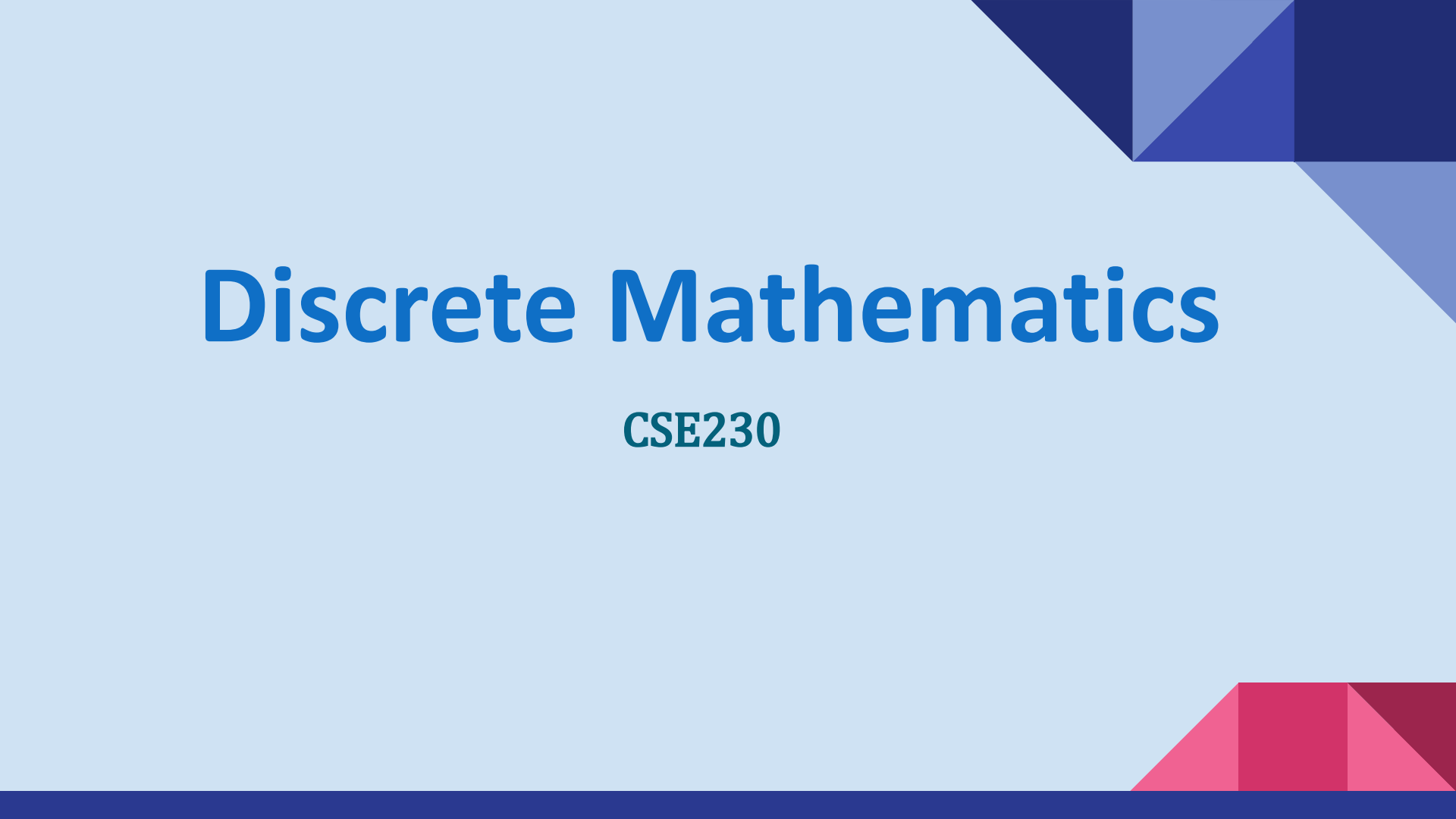


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Discrete Mathematics

CSE230

Discrete Mathematics and Its Applications

Kenneth H Rosen, New York: McGraw-Hill, 2019 (2003)

The book was sold more than 450K copies in US and is taught in more than 600 North American schools. It has been translated into Spanish, French, Portuguese, Greek, Chinese, Vietnamese and Korean.

This text is designed for a one- or two-term introductory discrete mathematics course taken by students in a wide variety of majors, including mathematics, computer science, and engineering. College algebra is the only explicit prerequisite, although a certain degree of mathematical maturity is needed to study discrete mathematics in a meaningful way.

Goals of a DM course

1. Mathematical Reasoning: to understand and construct mathematical reasoning starting with mathematical logic, mathematical induction
2. Combinatorial Analysis: ability of enumerating objects, counting
3. Discrete Structures: abstract mathematical structures like sets, permutations, relations, graphs, trees, finite state machines
4. Algorithmic Thinking: Some problems are solved by the specification of an algorithm, verification that it works properly and analysis of resource requirements
5. DM has applications to computer science, [data networking](#), chemistry, biology, linguistics, geography, business and internet

CSE230 Topic Coverage

The course is designed with a focus on covering the first half of **Kenneth H. Rosen's textbook**. Studying the assigned textbook sections each week is **mandatory** for a student. Students should also utilize **practice problem-sets** to prepare for exams.

Specifically, this course will cover:

- ✓ Propositional Logic Basics (Chapter 1),
- ✓ Argumentative and Inductive Proofs (Chapter 1 and 5),
- ✓ Set Theory and Applications (Chapter 2),
- ✓ Series and Recurrence (Chapter 2 and 8),
- ✓ Number Theory and Applications (Chapter 4) and
- ✓ Basics of counting and combinatorics (Chapter 6).

Quiz Policy

There will be at least 5 quizzes over the duration of this course. Unless any unavoidable situation arises, there will be **one quiz every 3 classes** , taking some small time out of the lectures/class time. The final grade will be based on the **best N-2** quiz scores.

Reason for such frequent quizzes is the absence of assignments in the course. Regular quizzes are there to encouraged regular study habits in students. With the exception of extreme cases, there will be **no make-up quizzes for absentees** .

Assignment Policy

There will be at least 3 assignments over the duration of this course. The assignments are expected to be published/announced with a **deadline of one week**, requiring **handwritten solutions** of questions prepared by the faculty.

For scoring, all the assignments will be counted. Deadlines are to be treated strictly. However, a small extension may be allowed by the faculty, provided that the student had valid reasons and requested an extension before the deadline.

Course Resources

All resources:

- | | |
|------------------------------------|---|
| • Course Content Folder (Student): | http://tiny.cc/cse230fall25 |
| • Official Discord Server: | https://discord.gg/[] [] [] []
(Follow instructions from the bot after joining) |
| • Book Link: | https://drive.google.com/file/d/1apBrWjrTiOLrPpY-z1PxW8QQ-i5Pp6wI/view?usp=drive link |

Marks Distribution

Attendance	90%+ attendance = 5 marks	5
Assignments	Average of 3	10
Class work / Quizzes	Best N – 2, out of 5 (Minimum)	20
Midterm Exam	Converted from 50	30
Final Exam	Converted from 50	35
Bonus	3+3.5 bonus from midterm+final (capped)	6.5
TOTAL		100+6.5

There are some topics that are reserved as “self-study” topics. In both midterm and final exams, there will be **bonus questions** worth 5 marks that can be added to the respective midterm and final scores, without exceeding the total marks.